

PSG Institute of Technology and Applied Research

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Branch: PSG iTech
Department: CSBS
Batch: 2027
Degree: B.Tech CSBS

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2027_IV_CSBS_Machine Learning Lab

PSG iTech_2027_Machine Learning 3_COD

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Santa needs to develop a model to predict employee attrition based on certain features. He has a CSV dataset containing employee data, including features like 'Age', 'YearsAtCompany', 'MonthlyIncome', 'PerformanceRating', 'WorkLifeBalance', and 'YearsInCurrentRole', along with the target variable 'Attrition'.

Santa requires a Python program that prompts him for this dataset's filename. The program should load the data, split it into training and testing sets, train a Naive Bayes classifier using Gaussian, make predictions on the testing set, and finally, output the accuracy of the model and the classification report.

Answer

main.py

```
# Train the Naive Bayes classifier (using GaussianNB)
```

```
# Make predictions
```

```
# Calculate metrics
```

```
# Generate classification report
```

```
model = GaussianNB()
```

```
model.fit(X_train, y_train)
```

```
l = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, l)
```

```
classification_rep = classification_report(y_test, l, zero_division=0)
```

input.csv

EmployeeID,FirstName,LastName,Age,Gender,Department,Position,YearsAtCompany,MonthlyIncome,PerformanceRating,WorkLifeBalance,YearsInCurrentRole,Attrition

1,John,Doe,30,Male,Sales,Sales Representative,2,5000,3,3,1,No

2,Jane,Smith,35,Female,IT,Software Engineer,5,6000,4,4,2,No

3,Bob,Johnson,40,Male,Finance,Accountant,8,7000,3,3,3,Yes

4,Amy,Williams,28,Female,HR,HR Assistant,1,4500,4,2,1,No

5,Mark,Anderson,45,Male,Marketing,Marketing Manager,10,9000,4,3,5,Yes

6,Lisa,Miller,33,Female,IT,Data Scientist,6,7500,3,4,2,No

7,Chris,Brown,38,Male,Finance,Financial Analyst,7,8000,4,3,3,Yes

8,Emily,Davis,29,Female,Sales,Sales Manager,4,7000,3,2,2,No

9,Michael,Smith,42,Male,HR,HR Manager,12,9500,4,4,6,Yes

10,Sarah,Johnson,31,Female,Marketing,Marketing Coordinator,3,5500,3,3,1,No

11,Daniel,White,36,Male,IT,IT Manager,9,8500,4,4,4,Yes

12,Karen,Jones,32,Female,Sales,Sales Representative,2,5000,3,3,1,No

13,Adam,Wilson,37,Male,Finance,Accountant,8,7000,3,3,3,Yes

14,Olivia,Thomas,27,Female,HR,HR Assistant,1,4500,4,2,1,No

23,Joshua,Walker,36,Male,Finance,Accountant,8,7000,3,3,3,Yes

24,Mia,Jackson,33,Female,HR,HR Assistant,1,4500,4,2,1,No

25,Brian,Evans,40,Male,Marketing,Marketing Manager,10,9000,4,3,5,Yes

input1.csv

EmployeeID,FirstName,LastName,Age,Gender,Department,Position,YearsAtCompany,MonthlyIncome,PerformanceRating,WorkLifeBalance,YearsInCurrentRole,Attrition

26,Nathan,Robinson,29,Male,IT,Software Engineer,4,6200,4,4,3,No
27,Amanda,Allen,34,Female,Sales,Sales Representative,3,5300,3,3,2,No
28,Ethan,Wright,39,Male,Finance,Financial Analyst,9,7800,4,3,4,Yes
29,Hannah,King,31,Female,HR,HR Manager,6,8600,4,4,2,No
30,Christopher,Lewis,37,Male,Marketing,Marketing Coordinator,5,5700,3,3,2,No
31,Lauren,Scott,28,Female,IT,Data Scientist,2,7200,4,4,1,No
32,Justin,Hall,33,Male,Finance,Accountant,7,7400,3,3,3,No
33,Katie,Young,30,Female,Sales,Sales Manager,4,6800,3,2,2,No
34,Brandon,Harris,36,Male,IT,IT Manager,8,8000,4,4,4,Yes
35,Elizabeth,Clark,29,Female,Marketing,Marketing Manager,6,8800,4,3,3,No
36,Tyler,Ward,32,Male,Finance,Financial Analyst,5,7600,4,3,2,No
37,Samantha,Turner,27,Female,HR,HR Assistant,1,4900,3,2,1,No
38,Matthew,Moore,35,Male,IT,Software Engineer,6,6500,3,4,2,No
39,Ashley,Parker,31,Female,Sales,Sales Representative,3,5200,3,3,2,No
40,David,Young,38,Male,Finance,Accountant,9,7900,3,3,3,Yes
41,Jessica,Kelly,30,Female,HR,HR Assistant,2,5100,3,2,1,No
42,Ryan,Bailey,34,Male,Marketing,Marketing Coordinator,5,5800,3,3,2,No
43,Stephanie,Cook,28,Female,IT,Data Scientist,3,7100,4,4,1,No
44,Andrew,Bennett,33,Male,Finance,Financial Analyst,8,7700,4,3,3,No
45,Megan,Richardson,29,Female,Sales,Sales Manager,4,6900,3,2,2,No
46,Jason,Cooper,36,Male,IT,IT Manager,9,8100,4,4,4,Yes
47,Heather,Walker,31,Female,Marketing,Marketing Manager,7,8700,4,3,3,No
48,William,Hall,32,Male,Finance,Financial Analyst,6,7500,4,3,2,No
49,Christina,Young,27,Female,HR,HR Assistant,1,4800,3,2,1,No
50,Eric,Adams,35,Male,IT,Software Engineer,7,6700,3,4,2,No

input2.csv

EmployeeID,FirstName,LastName,Age,Gender,Department,Position,YearsAtCompany,MonthlyIncome,PerformanceRating,WorkLifeBalance,YearsInCurrentRole,Attrition

51,Michelle,Hill,29,Female,Finance,Accountant,4,7200,3,3,2,No
52,Steven,Wright,34,Male,Sales,Sales Representative,3,5600,3,3,2,No
53,Sarah,Baker,39,Female,IT,Software Engineer,8,7800,4,4,3,No
54,Adam,Morris,31,Male,Marketing,Marketing Coordinator,5,5900,3,3,2,No
55,Emily,Campbell,28,Female,Finance,Financial Analyst,2,6500,4,3,1,No
56,Christopher,Hill,33,Male,HR,HR Assistant,6,7300,4,3,2,No
57,Samantha,Watson,30,Female,IT,Data Scientist,3,7100,4,4,1,No
58,Matthew,Turner,35,Male,Finance,Accountant,7,7500,3,3,3,No
59,Alexis,Adams,29,Female,Sales,Sales Manager,5,6800,3,2,2,No
60,Nathan,Reed,34,Male,IT,IT Manager,9,8000,4,4,4,Yes
61,Amanda,Wright,31,Female,Marketing,Marketing Manager,6,8500,4,3,3,No
62,Andrew,Phillips,28,Male,Finance,Financial Analyst,3,6700,4,3,2,No

63,Lauren,Cook,33,Female,HR,HR Assistant,2,5200,3,2,1,No
64,Justin,Bennett,30,Male,IT,Software Engineer,4,7000,3,4,2,No
65,Kayla,Ward,35,Female,Sales,Sales Representative,5,5400,3,3,2,No
66,Daniel,Young,32,Male,Marketing,Marketing Coordinator,7,5900,3,3,2,No
67,Hailey,James,27,Female,Finance,Accountant,4,7100,4,3,2,No
68,Michael,King,36,Male,HR,HR Manager,8,7800,4,4,2,No
56,Christopher,Hill,33,Male,HR,HR Assistant,6,7300,4,3,2,No
57,Samantha,Watson,30,Female,IT,Data Scientist,3,7100,4,4,1,No
58,Matthew,Turner,35,Male,Finance,Accountant,7,7500,3,3,3,No
59,Alexis,Adams,29,Female,Sales,Sales Manager,5,6800,3,2,2,No
60,Nathan,Reed,34,Male,IT,IT Manager,9,8000,4,4,4,Yes
61,Amanda,Wright,31,Female,Marketing,Marketing Manager,6,8500,4,3,3,No
62,Andrew,Phillips,28,Male,Finance,Financial Analyst,3,6700,4,3,2,No
63,Lauren,Cook,33,Female,HR,HR Assistant,2,5200,3,2,1,No
64,Justin,Bennett,30,Male,IT,Software Engineer,4,7000,3,4,2,No

input3.csv

EmployeeID,FirstName,LastName,Age,Gender,Department,Position,YearsAtComp
any,MonthlyIncome,PerformanceRating,WorkLifeBalance,YearsInCurrentRole,Attri
tion

51,Michelle,Hill,29,Female,Finance,Accountant,4,7200,3,3,2,No
52,Steven,Wright,34,Male,Sales,Sales Representative,3,5600,3,3,2,No
53,Sarah,Baker,39,Female,IT,Software Engineer,8,7800,4,4,3,No
54,Adam,Morris,31,Male,Marketing,Marketing Coordinator,5,5900,3,3,2,No
55,Emily,Campbell,28,Female,Finance,Financial Analyst,2,6500,4,3,1,No
56,Christopher,Hill,33,Male,HR,HR Assistant,6,7300,4,3,2,No
57,Samantha,Watson,30,Female,IT,Data Scientist,3,7100,4,4,1,No
58,Matthew,Turner,35,Male,Finance,Accountant,7,7500,3,3,3,No
59,Alexis,Adams,29,Female,Sales,Sales Manager,5,6800,3,2,2,No
60,Nathan,Reed,34,Male,IT,IT Manager,9,8000,4,4,4,Yes
61,Amanda,Wright,31,Female,Marketing,Marketing Manager,6,8500,4,3,3,No
62,Andrew,Phillips,28,Male,Finance,Financial Analyst,3,6700,4,3,2,No
63,Lauren,Cook,33,Female,HR,HR Assistant,2,5200,3,2,1,No
64,Justin,Bennett,30,Male,IT,Software Engineer,4,7000,3,4,2,No
65,Kayla,Ward,35,Female,Sales,Sales Representative,5,5400,3,3,2,No
66,Daniel,Young,32,Male,Marketing,Marketing Coordinator,7,5900,3,3,2,No
67,Hailey,James,27,Female,Finance,Accountant,4,7100,4,3,2,No
68,Michael,King,36,Male,HR,HR Manager,8,7800,4,4,2,No
69,Elizabeth,Roberts,31,Female,IT,Data Scientist,5,7200,4,4,2,No
70,Kevin,Johnson,29,Male,Finance,Financial Analyst,6,7300,4,3,3,No
71,Katherine,Lopez,34,Female,Sales,Sales Manager,7,7600,3,2,2,No
72,Brandon,Harris,30,Male,IT,IT Manager,9,8200,4,4,4,Yes

73,Jessica,Kelly,33,Female,Marketing,Marketing Manager,6,8800,4,3,3,No
74,Aaron,Wood,28,Male,Finance,Accountant,3,6700,4,3,2,No
75,Sophia,Young,35,Female,HR,HR Assistant,2,5300,3,2,1,No
76,Robert,Bell,32,Male,IT,Software Engineer,4,6900,3,4,2,No
77,Alyssa,Bailey,27,Female,Sales,Sales Representative,5,5500,3,3,2,No
78,Joseph,Ramirez,31,Male,Marketing,Marketing Coordinator,7,5900,3,3,2,No
79,Samantha,Watson,34,Female,Finance,Financial Analyst,8,7700,4,3,3,No

input4.csv

EmployeeID,FirstName,LastName,Age,Gender,Department,Position,YearsAtComp
any,MonthlyIncome,PerformanceRating,WorkLifeBalance,YearsInCurrentRole,Attri
tion

1,John,Doe,30,Male,Sales,Sales Representative,2,5000,3,3,1,No
2,Jane,Smith,35,Female,IT,Software Engineer,5,6000,4,4,2,No
3,Bob,Johnson,40,Male,Finance,Accountant,8,7000,3,3,3,Yes
4,Amy,Williams,28,Female,HR,HR Assistant,1,4500,4,2,1,No
5,Mark,Anderson,45,Male,Marketing,Marketing Manager,10,9000,4,3,5,Yes
6,Lisa,Miller,33,Female,IT,Data Scientist,6,7500,3,4,2,No
7,Chris,Brown,38,Male,Finance,Financial Analyst,7,8000,4,3,3,Yes
8,Emily,Davis,29,Female,Sales,Sales Manager,4,7000,3,2,2,No
9,Michael,Smith,42,Male,HR,HR Manager,12,9500,4,4,6,Yes
10,Sarah,Johnson,31,Female,Marketing,Marketing Coordinator,3,5500,3,3,1,No
11,Daniel,White,36,Male,IT,IT Manager,9,8500,4,4,4,Yes
56,Christopher,Hill,33,Male,HR,HR Assistant,6,7300,4,3,2,No
57,Samantha,Watson,30,Female,IT,Data Scientist,3,7100,4,4,1,No
58,Matthew,Turner,35,Male,Finance,Accountant,7,7500,3,3,3,No
59,Alexis,Adams,29,Female,Sales,Sales Manager,5,6800,3,2,2,No
60,Nathan,Reed,34,Male,IT,IT Manager,9,8000,4,4,4,Yes
61,Amanda,Wright,31,Female,Marketing,Marketing Manager,6,8500,4,3,3,No
62,Andrew,Phillips,28,Male,Finance,Financial Analyst,3,6700,4,3,2,No
63,Lauren,Cook,33,Female,HR,HR Assistant,2,5200,3,2,1,No
64,Justin,Bennett,30,Male,IT,Software Engineer,4,7000,3,4,2,No
65,Kayla,Ward,35,Female,Sales,Sales Representative,5,5400,3,3,2,No
66,Daniel,Young,32,Male,Marketing,Marketing Coordinator,7,5900,3,3,2,No
67,Hailey,James,27,Female,Finance,Accountant,4,7100,4,3,2,No
59,Alexis,Adams,29,Female,Sales,Sales Manager,5,6800,3,2,2,No
60,Nathan,Reed,34,Male,IT,IT Manager,9,8000,4,4,4,Yes
61,Amanda,Wright,31,Female,Marketing,Marketing Manager,6,8500,4,3,3,No
62,Andrew,Phillips,28,Male,Finance,Financial Analyst,3,6700,4,3,2,No
63,Lauren,Cook,33,Female,HR,HR Assistant,2,5200,3,2,1,No
64,Justin,Bennett,30,Male,IT,Software Engineer,4,7000,3,4,2,No

Status : Correct

Marks : 10/10

2. Problem Statement

Teju is working on a project where she needs to predict payment methods based on certain features. She has a CSV dataset containing features like 'Quantity', 'PricePerUnit', and 'TotalPrice', along with corresponding payment methods.

Teju requires a Python program that prompts her for the filename containing this dataset. The program should load the data, split it into training and testing sets, train a Multinomial Naive Bayes classifier, make predictions on the testing set, and finally, output the accuracy of the model.

Answer

main.py

```
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
from sklearn.model_selection import train_test_split
import pandas as pd
import numpy as np
import sys
import os
```

```
file = input()
path = os.path.join(sys.path[0], file)
data = pd.read_csv(path)
```

```
x = data.iloc[:, 5:8]
y = data.iloc[:, -2]
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3,
random_state=42)
```

```
model = MultinomialNB()
model.fit(x_train, y_train)
```

```
l = model.predict(x_test)
```

```
s = accuracy_score(y_test, l)
```

```
print(f"Accuracy: {s:.2f}")
```

input.csv

TransactionID,CustomerName,ProductID,ProductName,Category,Quantity,PricePerUnit,TotalPrice,PaymentMethod,TransactionDate

1,John Doe,101,Smartphone,Electronics,2,500,1000,Credit Card,2024-03-01
2,Jane Smith,102,Laptop,Electronics,1,800,800,Cash,2024-03-02
3,Bob Johnson,103,Tablet,Electronics,3,300,900,Debit Card,2024-03-03
4,Amy Williams,104,Headphones,Electronics,2,50,100,PayPal,2024-03-04
5,Mark Anderson,105,Smartwatch,Electronics,1,200,200,Credit Card,2024-03-05
6,Lisa Miller,106,Desk,HomeFurniture,1,150,150,Cash,2024-03-06
7,Chris Brown,107,Chair,HomeFurniture,4,80,320,Debit Card,2024-03-07
8,Emily Davis,108,Sofa,HomeFurniture,1,400,400,PayPal,2024-03-08
9,Michael Smith,109,Television,Electronics,2,1000,2000,Credit Card,2024-03-09
10,Sarah Johnson,110,Refrigerator,Appliances,1,700,700,Cash,2024-03-10
11,Daniel White,111,Blender,Appliances,2,30,60,Debit Card,2024-03-11
12,Karen Jones,112,VacuumCleaner,Appliances,1,150,150,PayPal,2024-03-12
13,Adam Wilson,113,CoffeeMaker,Appliances,3,50,150,Credit Card,2024-03-13
14,Olivia Thomas,114,Toaster,Appliances,2,20,40,Cash,2024-03-14
15,Benjamin Martinez,115,Bed,HomeFurniture,1,300,300,Debit Card,2024-03-15
16,Ava Garcia,116,DiningTable,HomeFurniture,1,250,250,PayPal,2024-03-16
17,James Rodriguez,117,Bookshelf,HomeFurniture,3,100,300,Credit Card,2024-03-17
18,Emma Lee,118,GamingConsole,Electronics,1,400,400,Cash,2024-03-18
19,Lucas Brown,119,Speaker,Electronics,2,100,200,Debit Card,2024-03-19
20,Isabella Martinez,120,Microwave,Appliances,1,80,80,PayPal,2024-03-20

input1.csv

TransactionID,CustomerName,ProductID,ProductName,Category,Quantity,PricePerUnit,TotalPrice,PaymentMethod,TransactionDate

21,Emily Johnson,121,Headphones,Electronics,2,70,140,PayPal,2024-04-01
22,William Smith,122,Smartwatch,Electronics,1,220,220,Credit Card,2024-04-02
23,Victoria Anderson,123,Desk,HomeFurniture,1,180,180,Cash,2024-04-03
24,Mason Lee,124,Chair,HomeFurniture,4,90,360,Debit Card,2024-04-04
25,Ella Hernandez,125,Sofa,HomeFurniture,1,450,450,PayPal,2024-04-05
26,Aiden Garcia,126,Television,Electronics,2,1050,2100,Credit Card,2024-04-06
27,Mia Martinez,127,Refrigerator,Appliances,1,720,720,Cash,2024-04-07
28,Alexander Johnson,128,Blender,Appliances,2,40,80,Debit Card,2024-04-08
29,Scarlett Miller,129,VacuumCleaner,Appliances,1,180,180,PayPal,2024-04-09
30,Leo Brown,130,CoffeeMaker,Appliances,3,60,180,Credit Card,2024-04-10

input2.csv

TransactionID,CustomerName,ProductID,ProductName,Category,Quantity,PricePerUnit,TotalPrice,PaymentMethod,TransactionDate

31,Lily Wilson,131,Toaster,Appliances,2,30,60,Cash,2024-04-11
32,Ethan Thomas,132,Bed,HomeFurniture,1,350,350,Debit Card,2024-04-12
33,Chloe Garcia,133,DiningTable,HomeFurniture,1,300,300,PayPal,2024-04-13
34,Daniel Rodriguez,134,Bookshelf,HomeFurniture,3,120,360,Credit Card,2024-04-14
35,Ava Hernandez,135,GamingConsole,Electronics,1,420,420,Cash,2024-04-15
36,Lucas Lopez,136,Speaker,Electronics,2,120,240,Debit Card,2024-04-16
37,Sophia Brown,137,Microwave,Appliances,1,90,90,PayPal,2024-04-17
38,William Perez,138,Smartphone,Electronics,2,550,1100,Credit Card,2024-04-18
39,Olivia Scott,139,Laptop,Electronics,1,850,850,Cash,2024-04-19
40,Noah Gonzalez,140,Tablet,Electronics,3,320,960,Debit Card,2024-04-20
41,Oliver Rodriguez,141,Desk,HomeFurniture,1,160,160,PayPal,2024-05-01
42,Amelia Martinez,142,Chair,HomeFurniture,3,80,240,Credit Card,2024-05-02
43,Ethan Johnson,143,Sofa,HomeFurniture,1,410,410,Cash,2024-05-03
44,Ava Garcia,144,Television,Electronics,2,990,1980,Debit Card,2024-05-04
45,Isabella Smith,145,Refrigerator,Appliances,1,720,720,PayPal,2024-05-05

input3.csv

TransactionID,CustomerName,ProductID,ProductName,Category,Quantity,PricePerUnit,TotalPrice,PaymentMethod,TransactionDate

46,Liam Brown,146,Blender,Appliances,2,45,90,Credit Card,2024-05-06
47,Sophia Davis,147,VacuumCleaner,Appliances,1,190,190,Cash,2024-05-07
48,James Wilson,148,CoffeeMaker,Appliances,3,55,165,Debit Card,2024-05-08
49,Charlotte Thomas,149,Toaster,Appliances,2,35,70,PayPal,2024-05-09
50,Noah Taylor,150,Bed,HomeFurniture,1,320,320,Credit Card,2024-05-10
51,Emily Garcia,151,DiningTable,HomeFurniture,1,280,280,Cash,2024-05-11
52,Lucas Johnson,152,Bookshelf,HomeFurniture,3,110,330,Debit Card,2024-05-12
53,Olivia Anderson,153,GamingConsole,Electronics,1,420,420,PayPal,2024-05-13
54,Lily Miller,154,Speaker,Electronics,2,130,260,Credit Card,2024-05-14
55,Ethan Martinez,155,Microwave,Appliances,1,95,95,Cash,2024-05-15
56,Amelia Hernandez,156,Smartphone,Electronics,2,570,1140,Debit Card,2024-05-16
57,William Perez,157,Laptop,Electronics,1,870,870,PayPal,2024-05-17
58,Sophia Brown,158,Tablet,Electronics,3,350,1050,Credit Card,2024-05-18
59,Lucas Lopez,159,Desk,HomeFurniture,1,180,180,Cash,2024-05-19
60,Ava Hernandez,160,Chair,HomeFurniture,3,90,270,Debit Card,2024-05-20

61,Noah Gonzalez,161,Sofa,HomeFurniture,1,430,430,PayPal,2024-05-21
62,Olivia Scott,162,Television,Electronics,2,1010,2020,Credit Card,2024-05-22
63,Oliver Rodriguez,163,Refrigerator,Appliances,1,740,740,Cash,2024-05-23
64,Amelia Martinez,164,Blender,Appliances,2,50,100,Debit Card,2024-05-24
65,Ethan Johnson,165,VacuumCleaner,Appliances,1,200,200,PayPal,2024-05-25
66,Ava Garcia,166,CoffeeMaker,Appliances,3,60,180,Credit Card,2024-05-26
67,Isabella Smith,167,Toaster,Appliances,2,40,80,Cash,2024-05-27
68,Liam Brown,168,Bed,HomeFurniture,1,350,350,Debit Card,2024-05-28
69,Sophia Davis,169,DiningTable,HomeFurniture,1,300,300,PayPal,2024-05-29
70,James Wilson,170,Bookshelf,HomeFurniture,3,120,360,Credit Card,2024-05-30

input4.csv

TransactionID,CustomerName,ProductID,ProductName,Category,Quantity,PricePerUnit,TotalPrice,PaymentMethod,TransactionDate

71, Lucas Johnson,171,GamingConsole,Electronics,1,430,430,Credit Card,2024-06-01
72,Olivia Anderson,172,Speaker,Electronics,2,130,260,PayPal,2024-06-02
73,Lily Miller,173,Microwave,Appliances,1,95,95,Cash,2024-06-03
74,Ethan Martinez,174,Smartphone,Electronics,2,570,1140,Debit Card,2024-06-04
75,William Perez,175,Laptop,Electronics,1,870,870,Credit Card,2024-06-05
76,Sophia Brown,176,Tablet,Electronics,3,350,1050,PayPal,2024-06-06
77, Lucas Lopez,177,Desk,HomeFurniture,1,180,180,Cash,2024-06-07
78,Ava Hernandez,178,Chair,HomeFurniture,3,90,270,Debit Card,2024-06-08
79,Noah Gonzalez,179,Sofa,HomeFurniture,1,430,430,PayPal,2024-06-09
80,Olivia Scott,180,Television,Electronics,2,1010,2020,Credit Card,2024-06-10
81,Oliver Rodriguez,181,Refrigerator,Appliances,1,740,740,Cash,2024-06-11
82,Amelia Martinez,182,Blender,Appliances,2,50,100,Debit Card,2024-06-12
83,Ethan Johnson,183,VacuumCleaner,Appliances,1,200,200,PayPal,2024-06-13
84,Ava Garcia,184,CoffeeMaker,Appliances,3,60,180,Credit Card,2024-06-14
85,Isabella Smith,185,Toaster,Appliances,2,40,80,Cash,2024-06-15
86,Liam Brown,186,Bed,HomeFurniture,1,350,350,Debit Card,2024-06-16
87,Sophia Davis,187,DiningTable,HomeFurniture,1,300,300,PayPal,2024-06-17
88,James Wilson,188,Bookshelf,HomeFurniture,3,120,360,Credit Card,2024-06-18
89,Charlotte Thomas,189,GamingConsole,Electronics,1,420,420,Cash,2024-06-19
90,Oliver Clark,190,Speaker,Electronics,2,140,280,Debit Card,2024-06-20
91,Lily Hernandez,191,Microwave,Appliances,1,90,90,PayPal,2024-06-21
92,Ethan Brown,192,Smartphone,Electronics,2,600,1200,Credit Card,2024-06-22
93,William Garcia,193,Laptop,Electronics,1,850,850,Cash,2024-06-23
94,Sophia White,194,Tablet,Electronics,3,400,1200,Debit Card,2024-06-24

95, Lucas Thompson, 195, Desk, HomeFurniture, 1, 200, 200, PayPal, 2024-06-25
96, Ava Rodriguez, 196, Chair, HomeFurniture, 3, 100, 300, Credit Card, 2024-06-26
97, Noah Perez, 197, Sofa, HomeFurniture, 1, 500, 500, Cash, 2024-06-27
98, Olivia Martinez, 198, Television, Electronics, 2, 1000, 2000, Debit Card, 2024-06-28
99, Oliver Lopez, 199, Refrigerator, Appliances, 1, 720, 720, PayPal, 2024-06-29
100, Amelia Thompson, 200, Blender, Appliances, 2, 50, 100, Credit Card, 2024-06-30
101, Emma Hernandez, 201, VacuumCleaner, Appliances, 1, 200, 200, Cash, 2024-06-31
102, Liam Johnson, 202, CoffeeMaker, Appliances, 3, 70, 210, Debit Card, 2024-06-32
103, Sophia Davis, 203, Toaster, Appliances, 2, 40, 80, PayPal, 2024-06-33
104, James Wilson, 204, Bed, HomeFurniture, 1, 300, 300, Credit Card, 2024-06-34
105, Charlotte Thomas, 205, DiningTable, HomeFurniture, 1, 250, 250, Cash, 2024-06-35
106, Oliver Clark, 206, Bookshelf, HomeFurniture, 3, 110, 330, Debit Card, 2024-06-36

Status : Correct

Marks : 10/10

3. Problem Statement

Sharmi is working on a project where she needs to predict labels based on certain features. She has a CSV dataset containing features like 'Price', 'UnitsSold', and 'Profit', along with corresponding labels. Sharmi requires a Python program that prompts her for the filename containing this dataset.

The program should load the data, split it into training and testing sets, train a Gaussian Naive Bayes classifier, make predictions on the testing set, and finally, output the accuracy of the model and the confusion matrix.

Answer

main.py

```
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
import os
import sys
import pandas as pd

file = input()
path = os.path.join(sys.path[0], file)
```

```
data = pd.read_csv(path)
```

```
x = data[['Price', 'UnitsSold', 'Profit']]  
y = data.iloc[:, -1]
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3,  
random_state=42)
```

```
model = GaussianNB()  
model.fit(x_train, y_train)
```

```
l = model.predict(x_test)  
s = accuracy_score(y_test, l)  
a = confusion_matrix(y_test, l)  
print(f"Accuracy: {s:.2f}\n")  
print("Confusion Matrix:\n", a)
```

sample.csv

ProductID	ProductName	Category	Price	UnitsSold	Profit	Label
101	Smartphone	Electronics	500	100	25000	HighDemand
102	Laptop	Electronics	800	80	32000	HighDemand
103	Tablet	Electronics	300	150	15000	MediumDemand
104	Headphones	Electronics	50	200	8000	MediumDemand
105	Smartwatch	Electronics	200	120	10000	MediumDemand
106	Desk	HomeFurniture	150	90	8000	LowDemand
107	Chair	HomeFurniture	80	120	6000	LowDemand
108	Sofa	HomeFurniture	400	50	15000	LowDemand
109	Television	Electronics	1000	60	40000	HighDemand
110	Refrigerator	Appliances	700	40	20000	HighDemand
111	Blender	Appliances	30	150	4000	MediumDemand
112	VacuumCleaner	Appliances	150	80	10000	MediumDemand
113	CoffeeMaker	Appliances	50	100	5000	MediumDemand
114	Toaster	Appliances	20	120	2000	LowDemand
115	Bed	HomeFurniture	300	70	12000	LowDemand
116	DiningTable	HomeFurniture	250	60	10000	LowDemand
117	Bookshelf	HomeFurniture	100	100	6000	LowDemand
118	GamingConsole	Electronics	400	70	18000	HighDemand
119	Speaker	Electronics	100	150	12000	MediumDemand
120	Microwave	Appliances	80	80	5000	LowDemand

sample1.csv

ProductID,ProductName,Category,Price,UnitsSold,Profit,Label
121,DigitalCamera,Electronics,300,50,10000,MediumDemand
122,PortableSpeaker,Electronics,80,200,8000,HighDemand
123,OfficeChair,HomeFurniture,120,90,6000,LowDemand
124,LEDTelevision,Electronics,1200,30,25000,HighDemand
125,SlowCooker,Appliances,60,100,5000,MediumDemand
126,CoffeeTable,HomeFurniture,200,80,7000,MediumDemand
127,HomeTheaterSystem,Electronics,600,40,18000,HighDemand
128,ToasterOven,Appliances,50,120,4000,MediumDemand
129,WirelessRouter,Electronics,100,150,10000,MediumDemand
130,ReclinerChair,HomeFurniture,350,60,15000,LowDemand
131,SmartSpeaker,Electronics,150,120,9000,MediumDemand
132,MiniFridge,Appliances,120,70,6000,LowDemand
133,ComputerDesk,HomeFurniture,100,100,8000,LowDemand
134,StandMixers,Appliances,200,50,8000,MediumDemand
135,BedsideTable,HomeFurniture,80,150,4000,MediumDemand
136,SmartHomeHub,Electronics,250,80,12000,MediumDemand
137,BarStools,HomeFurniture,50,200,6000,HighDemand
138,WirelessEarbuds,Electronics,100,120,8000,HighDemand
139,SteamIron,Appliances,30,180,2000,LowDemand
140,WallShelves,HomeFurniture,40,250,3000,HighDemand

sample2.csv

ProductID,ProductName,Category,Price,UnitsSold,Profit,Label
141,WirelessHeadphones,Electronics,150,100,8000,HighDemand
142,SectionalSofa,HomeFurniture,800,30,15000,MediumDemand
143,AirPurifier,Appliances,200,50,6000,MediumDemand
144,SmartThermostat,Electronics,100,80,4000,LowDemand
145,BarbecueGrill,HomeFurniture,300,40,7000,MediumDemand
146,ElectricKettle,Appliances,40,120,2000,LowDemand
147,GamingChair,Electronics,250,70,8000,MediumDemand
148,BeanBagChair,HomeFurniture,70,150,3000,HighDemand
149,WirelessCharger,Electronics,30,200,4000,HighDemand
150,StandingDesk,HomeFurniture,400,60,12000,MediumDemand
151,RoboticVacuum,Appliances,300,40,10000,MediumDemand
152,Projector,Electronics,500,20,8000,LowDemand
153,OutdoorFurnitureSet,HomeFurniture,1000,10,15000,HighDemand
154,RiceCooker,Appliances,50,100,4000,MediumDemand
155,WallArt,HomeFurniture,50,150,2000,LowDemand
156,WirelessSecurityCamera,Electronics,120,80,6000,MediumDemand
157,SteamMop,Appliances,80,90,3000,LowDemand
158,BluetoothSpeaker,Electronics,70,120,5000,MediumDemand

159,CounterStools,HomeFurniture,80,100,4000,MediumDemand
160,MiniBlender,Appliances,30,150,2000,LowDemand
101,Smartphone,Electronics,500,100,25000,HighDemand
102,Laptop,Electronics,800,80,32000,HighDemand
103,Tablet,Electronics,300,150,15000,MediumDemand
104,Headphones,Electronics,50,200,8000,MediumDemand
105,Smartwatch,Electronics,200,120,10000,MediumDemand
106,Desk,HomeFurniture,150,90,8000,LowDemand
107,Chair,HomeFurniture,80,120,6000,LowDemand
108,Sofa,HomeFurniture,400,50,15000,LowDemand

sample3.csv

ProductID,ProductName,Category,Price,UnitsSold,Profit,Label
161,WirelessMouse,Electronics,40,150,2000,MediumDemand
162,EndTable,HomeFurniture,80,100,3000,MediumDemand
163,ElectricToothbrush,Appliances,50,120,4000,MediumDemand
164,SmartLock,Electronics,100,80,6000,MediumDemand
165,ConsoleTable,HomeFurniture,120,70,5000,MediumDemand
166,HandBlender,Appliances,30,150,2500,MediumDemand
167,GamingKeyboard,Electronics,60,100,4000,MediumDemand
168,DiningChair,HomeFurniture,70,120,3500,MediumDemand
169,PortableFan,Appliances,20,200,1500,MediumDemand
170,BluetoothHeadset,Electronics,50,130,3000,MediumDemand
171,StorageCabinet,HomeFurniture,150,60,8000,LowDemand
172,WaterDispenser,Appliances,100,90,5000,LowDemand
173,WirelessPrinter,Electronics,200,40,6000,LowDemand
174,CoffeeTableSet,HomeFurniture,200,50,9000,LowDemand
175,FoodProcessor,Appliances,80,110,4000,MediumDemand
176,LaptopStand,Electronics,30,180,2500,MediumDemand
177,CoatRack,HomeFurniture,40,160,2000,MediumDemand
178,PersonalBlender,Appliances,40,150,3000,MediumDemand
179,DesktopMonitor,Electronics,150,70,5000,MediumDemand
180,BedsideLamp,HomeFurniture,50,120,2500,MediumDemand
181,ElectricGrill,Appliances,100,80,4000,MediumDemand
182,SmartScale,Electronics,60,100,2000,MediumDemand
183,TVStand,HomeFurniture,100,90,4500,MediumDemand
184,HandheldVacuum,Appliances,80,110,3500,MediumDemand
185,ComputerMousePad,Electronics,10,200,1000,MediumDemand
186,Bookcase,HomeFurniture,120,60,6000,MediumDemand
187,Steamer,Appliances,50,130,2500,MediumDemand
188,WirelessRouter,Electronics,70,120,4000,MediumDemand
189,StorageOttoman,HomeFurniture,60,140,3000,MediumDemand

190,HandMixer,Appliances,40,150,2000,MediumDemand

sample4.csv

ProductID,ProductName,Category,Price,UnitsSold,Profit,Label

161,SmartTV,Electronics,700,50,35000,HighDemand

162,OfficeDesk,HomeFurniture,200,70,14000,MediumDemand

163,Blender,Appliances,40,150,3000,LowDemand

164,WirelessMouse,Electronics,30,200,4000,HighDemand

165,OutdoorSofa,HomeFurniture,500,30,12000,MediumDemand

166,HandheldVacuum,Appliances,50,100,5000,MediumDemand

167,GamingKeyboard,Electronics,100,120,8000,HighDemand

168,BeanBag,HomeFurniture,50,150,2000,LowDemand

169,USBFlashDrive,Electronics,20,250,3000,HighDemand

170,StandingFan,Appliances,30,180,4000,MediumDemand

171,Printer,Electronics,150,80,8000,MediumDemand

172,PatioTable,HomeFurniture,300,40,10000,MediumDemand

173,SlowJuicer,Appliances,80,90,6000,MediumDemand

174,WallClock,HomeFurniture,20,200,2000,HighDemand

175,WirelessKeyboard,Electronics,50,120,6000,MediumDemand

176,MiniFreezer,Appliances,100,70,5000,MediumDemand

177,SmartDoorbell,Electronics,120,80,7000,MediumDemand

178,ShoeRack,HomeFurniture,40,150,3000,LowDemand

179,PortableCharger,Electronics,30,200,4000,HighDemand

180,RobotVacuum,Appliances,200,50,10000,MediumDemand

181,LaptopStand,Electronics,25,250,4000,HighDemand

182,OutdoorDiningSet,HomeFurniture,800,20,16000,HighDemand

183,ElectricGriddle,Appliances,60,100,5000,MediumDemand

184,PhotoFrame,HomeFurniture,15,300,2000,LowDemand

185,WirelessEarphones,Electronics,80,120,6000,MediumDemand

186,Steamer,Appliances,40,150,3000,LowDemand

187,SmartScale,Electronics,50,120,6000,MediumDemand

188,EndTable,HomeFurniture,30,200,4000,HighDemand

189,HandBlender,Appliances,35,180,3000,MediumDemand

190,BluetoothHeadset,Electronics,60,100,5000,MediumDemand

191,ClothesDryingRack,HomeFurniture,25,250,4000,HighDemand

192,FoodProcessor,Appliances,70,90,6000,MediumDemand

193,DeskLamp,Electronics,40,150,2000,LowDemand

194,Nightstand,HomeFurniture,40,150,2000,LowDemand

195,IceCreamMaker,Appliances,80,90,7000,MediumDemand

196,SmartLightBulb,Electronics,25,300,4000,HighDemand

197,BarCart,HomeFurniture,100,70,5000,MediumDemand

198,ElectricKettle,Appliances,30,180,3000,MediumDemand

199,GamingMouse,Electronics,50,120,6000,MediumDemand
200,DeskOrganizer,HomeFurniture,20,200,4000,HighDemand

Status : Correct

Marks : 10/10

4. Problem Statement

Janani needs to develop a model for predicting labels based on certain environmental parameters. She has a dataset in CSV format containing columns for 'Temperature', 'Humidity', 'Wind_Speed', 'Pressure', and 'Cloud_Cover', along with corresponding labels.

Janani requires a Python program that prompts her for the filename containing this dataset. The program should load the data, split it into training and testing sets, train a Gaussian Naive Bayes classifier, make predictions on the testing set, and finally, output the accuracy of the model rounded to two decimal places and the confusion matrix.

Answer

main.py

```
import os
import sys
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.naive_bayes import GaussianNB

file = input()
path = os.path.join(sys.path[0], file)
data = pd.read_csv(path)
x, y = data.iloc[:, :-1], data.iloc[:, -1]
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3,
random_state=42)
model = GaussianNB()
model.fit(x_train, y_train)
l = model.predict(x_test)
s = accuracy_score(y_test, l)
print(f"Accuracy: {s:.2f}")
c = confusion_matrix(y_test, l)
```

```
print("Confusion Matrix:\n", c)
```

sample.csv

Temperature,Humidity,Wind_Speed,Pressure,Cloud_Cover,Label

28,75,10,1015,30,No

22,60,15,1010,20,Yes

30,80,8,1012,40,Yes

25,70,12,1018,25,No

27,85,18,1016,50,Yes

23,65,14,1014,15,No

29,78,9,1011,35,Yes

26,68,11,1017,22,No

24,72,16,1013,45,Yes

31,82,7,1010,28,Yes

20,58,13,1019,10,No

28,79,10,1014,32,Yes

21,63,17,1011,18,No

30,83,8,1016,38,Yes

25,71,12,1013,24,No

27,86,19,1015,48,Yes

22,64,15,1018,14,No

29,77,9,1012,33,Yes

26,67,11,1017,21,No

23,70,16,1014,43,Yes

32,81,7,1009,26,Yes

19,59,13,1019,12,No

28,80,10,1013,29,Yes

24,66,14,1016,16,No

26,84,18,1011,46,Yes

sample1.csv

Temperature,Humidity,Wind_Speed,Pressure,Cloud_Cover,Label

29,72,11,1014,28,No

23,65,16,1011,17,Yes

31,78,9,1013,35,Yes

26,69,12,1017,23,No

28,84,17,1015,47,Yes

22,63,15,1018,13,No

30,77,10,1012,31,Yes

25,67,13,1016,20,No

27,73,18,1010,40,Yes

32,80,8,1011,30,Yes
21,60,14,1019,10,No
31,76,10,1012,32,Yes
27,70,13,1017,21,No
22,61,16,1014,42,Yes
33,82,7,1010,27,Yes
20,59,13,1019,11,No
28,79,11,1013,30,Yes
25,68,14,1016,18,No
27,83,18,1011,45,Yes

sample2.csv

Temperature,Humidity,Wind_Speed,Pressure,Cloud_Cover,Label

30,75,10,1012,30,No
22,60,15,1010,20,Yes
29,80,8,1013,40,Yes
25,70,12,1018,25,No
27,85,18,1016,50,Yes
23,65,14,1014,15,No
29,78,9,1011,35,Yes
26,68,11,1017,22,No
24,72,16,1013,45,Yes
31,82,7,1010,28,Yes
20,58,13,1019,10,No
28,79,10,1014,32,Yes
21,63,17,1011,18,No
30,83,8,1016,38,Yes
25,71,12,1013,24,No
27,86,19,1015,48,Yes
22,64,15,1018,14,No
29,77,9,1012,33,Yes
26,67,11,1017,21,No
23,70,16,1014,43,Yes
32,81,7,1009,26,Yes
19,59,13,1019,12,No
28,80,10,1013,29,Yes
24,66,14,1016,16,No
26,84,18,1011,46,Yes
30,75,10,1012,30,No
22,60,15,1010,20,Yes
29,80,8,1013,40,Yes
25,70,12,1018,25,No

27,85,18,1016,50,Yes

23,65,14,1014,15,No

sample3.csv

Temperature,Humidity,Wind_Speed,Pressure,Cloud_Cover,Label

30,75,10,1012,30,No

22,60,15,1010,20,Yes

29,80,8,1013,40,Yes

25,70,12,1018,25,No

27,85,18,1016,50,Yes

23,65,14,1014,15,No

29,78,9,1011,35,Yes

26,68,11,1017,22,No

24,72,16,1013,45,Yes

31,82,7,1010,28,Yes

20,58,13,1019,10,No

28,79,10,1014,32,Yes

21,63,17,1011,18,No

30,83,8,1016,38,Yes

25,71,12,1013,24,No

27,86,19,1015,48,Yes

22,64,15,1018,14,No

29,77,9,1012,33,Yes

26,67,11,1017,21,No

23,70,16,1014,43,Yes

32,81,7,1009,26,Yes

19,59,13,1019,12,No

28,80,10,1013,29,Yes

24,66,14,1016,16,No

26,84,18,1011,46,Yes

30,75,10,1012,30,No

22,60,15,1010,20,Yes

29,80,8,1013,40,Yes

25,70,12,1018,25,No

27,85,18,1016,50,Yes

23,65,14,1014,15,No

29,78,9,1011,35,Yes

26,68,11,1017,22,No

24,72,16,1013,45,Yes

31,82,7,1010,28,Yes

20,58,13,1019,10,No

28,79,10,1014,32,Yes

21,63,17,1011,18,No
30,83,8,1016,38,Yes
25,71,12,1013,24,No

sample4.csv

Temperature,Humidity,Wind_Speed,Pressure,Cloud_Cover,Label

30,75,10,1012,30,No
22,60,15,1010,20,Yes
29,80,8,1013,40,Yes
25,70,12,1018,25,No
27,85,18,1016,50,Yes
23,65,14,1014,15,No
29,78,9,1011,35,Yes
26,68,11,1017,22,No
24,72,16,1013,45,Yes
31,82,7,1010,28,Yes
20,58,13,1019,10,No
28,79,10,1014,32,Yes
21,63,17,1011,18,No
30,83,8,1016,38,Yes
25,71,12,1013,24,No
27,86,19,1015,48,Yes
22,64,15,1018,14,No
29,77,9,1012,33,Yes
26,67,11,1017,21,No
23,70,16,1014,43,Yes
32,81,7,1009,26,Yes
19,59,13,1019,12,No
28,80,10,1013,29,Yes
24,66,14,1016,16,No
26,84,18,1011,46,Yes
30,75,10,1012,30,No
22,60,15,1010,20,Yes
29,80,8,1013,40,Yes
25,70,12,1018,25,No
27,85,18,1016,50,Yes
23,65,14,1014,15,No
29,78,9,1011,35,Yes
26,68,11,1017,22,No
24,72,16,1013,45,Yes
31,82,7,1010,28,Yes
20,58,13,1019,10,No

28,79,10,1014,32,Yes
21,63,17,1011,18,No
30,83,8,1016,38,Yes
25,71,12,1013,24,No
27,86,19,1015,48,Yes
22,64,15,1018,14,No
29,77,9,1012,33,Yes
26,67,11,1017,21,No
23,70,16,1014,43,Yes
32,81,7,1009,26,Yes
19,59,13,1019,12,No

Status : Correct

Marks : 10/10